Player Testing

1 Project Information

Game Title: Flap-and-Fight
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Build Version Tested: Version 2
Date(s) of Playtesting: April 8, 2026

2 Purpose of the Player Test

Playtest helps us developers refine the game we created and see it being played by actual players instead of us the developers who made the game. The specific purpose of this playtest are:

- Uncover hidden problems: Bugs, inconsistencies and glitches
- Gain unbiased feedback
- Enhance gameplay
- Improve user experience
- Clear precise action generation

3 Participants

There were three participants that playtested the game. The players were all fellow TRU software engineering students and fellow agile game development classmates. Most of them play games regularly and can easily understand game mechanics. A variety of skill levels were witnessed as they played the game.

Unfortunately the playtesters were not fully representative of the player base that the game should appeal to. The play testers were on the more experienced side of the skill spectrum and all of them were not new to playing the game.

4 Test Artifacts

Game build

- Version: Version 2 (Mid-sprint 5 prototype)
- Core Features included**:**
    - **Movement:** Vertical flap mechanics, gravity physics, and automatic horizontal progression.
    - **Combat:** Basic projectile attack (bullet) and stationary enemy types.
    - **Obstacles:** Procedurally generated pipe hazards with randomized vertical positioning.
    - **UI/UX:** Start/Entry screen with hovering idle, HUD (Health/Hearts display), Score tracking, and a Game Over screen.

- **Systems:** Difficulty scaling (speed increases over time) and Power-ups (Hearts and Shields).

Platform

- **Engine**: Unity 2D
- **Hardware:** Tested on PC/Laptop (Windows and Mac compatibility verified via Git LFS).
- **Input Method:** Keyboard (Mouse Click for flapping and space key projectile firing).

Known Limitations & Bugs

- **Incomplete Mechanics:** The "Hold-to-Glide" and "Aggressive Dive Enemy" features (PB-09 and PB-17) are currently in development and were not fully functional for this specific build.
- **Visual Feedback:** Hit feedback (particles/vfx) is still being implemented, so some players may find it difficult to tell if a bullet has registered a hit on an enemy.
- **Collision Tuning:** While pipes cause instant death, player health is currently tied only to enemy collisions and power-ups; this logic is still being balanced for fairness.
- **Asset Placeholders:** Some sprites are temporary free-source assets and may not reflect the final art style.

5 Tasks and Test Script

Directed tasks: ensure players interact with specific mechanics required to play.

- Task1: Initialization
  - Launch the game and wait until entry screen appears
  - About 10 seconds
- Task 2: Navigation
  - Attempt to pass through 5-10 obstacles
- Task 3: Combat
  - Use attack mechanism (bullet) to destroy at least 2 enemies
- Task 4: Sustainability
  - Take damage and and get restore health from health powerup
- Task 5: Termination
  - Allow collision to pipe to verify gameover

Open-ended tasks:

- Free play:
  - Play game for 5-10 minutes
  - Focus on the feel of the game and how they flap with the gravity setup

- Difficulty Assessment:
  - Continue playing until they notice difficulty increasing and take note of whey players struggle.
- Mechanical Exploration:
  - Combine movement and attack simultaneously and see if the combination is intuitive.
  - See if they get “stuck” figuring out to flap and shoot at the same time.
- Visual clarity:
  - Observer heart score while playing and not if they are able to keep track of their health.

Test script/ prompt for players:

“This is our game, Flap-and-Fight. Today, we want to see how the new movement, combat and power-up systems feel. Please start at the entry screen. Left click is used to flap and space is used to shoot at the enemy. First, try to survive long enough to see the game speed up. Then, try using your projectiles to destroy enemies in your path. Don't worry about dying since this game is infinite there will not be levels. We want to see how you handle the obstacles and if the game's difficulty feels rewarding or frustrating.”

6 Metrics Collected

Quantitative Metrics

These data points provide objective evidence of game balance and technical stability.

- **Average Survival Time:**
  - Ranges from 5 seconds to 20 seconds in the first 2 plays testers ever played. A maximum score of 10 points was recorded.
  - After the 3rd try of the game players got more of a handle and we saw a range of 10–43 points since players were getting used to the coordination needed to play this game.
- **Death Distribution:**
  - Instant death was rare but seen at the immediate beginning of the game.
  - Most deaths were from pipes, about 90%.
  - Initially, players almost 100% of the time avoided the enemies because it was easier.
  - As testing progressed more and more players started using the bullet feature and the death from pipe and enemy became more of 70% and 30%.
  - Almost 100% of the time, players died from hitting pipes and not losing hearts.

- **Combat Efficiency:**
    - Bullets fired to enemies hit about 4 out of 10.
- **Power-up Acquisition:**
    - Number of hearts collected = average of 4 per session
    - Number of shields collected = N/A
- **High Score Tracking:**
    - Highest score achieved in the playtest = 42

Qualitative Metrics

These data points capture the player's subjective experience, providing insight into the "feel" of the game.

- **Think-Aloud Observations:**
    - Flapping and shooting was difficult in terms of hand eye coordination.
    - "This mechanism is really difficult at first."
    - "Pipe spacing could be bigger."
    - "Gravity could be lowered or more fine tuned with mass."
- **Control Responsiveness:**
    - The first comment was "the bird felt a little heavy and the gravity was a little big."
- **UI/HUD Clarity:**
    - Players knew about them having hearts since it is big and clear on the screen.
- **Difficulty Scaling Perception:**
    - Smooth speed increment and not noticeable.
    - Another reason it isn't noticeable was because most didn't play to see this progression.
- **Visual Feedback Satisfaction:**
    - Got a comment to add visual or audio feedback to when killing enemies or bullet colliding.

7 Observations and Results

Player Behavior Patterns

- **Skill Acquisition Curve:**
    - Players initially struggled with the high coordination required for simultaneous flapping and shooting, with survival times starting as low as 5 seconds.
    - However, a significant learning curve was observed; by the third attempt, players improved their coordination, increasing their scores from a 10-point maximum to

a high of 43 points.

- **Avoidance Tactics:**
  - In early sessions, players exhibited a 100% avoidance rate toward enemies, preferring to focus purely on navigation.
  - This behavior shifted as they gained confidence, eventually engaging in combat and balancing death causes between pipes (70%) and enemies (30%).
- **Power-up Utilization:**
  - Players successfully integrated sustainability mechanics into their playstyle, averaging 4 heart collections per session.

Confusion Points and Friction

- **Lethality Imbalance:**
  - A recurring pattern showed that almost 100% of deaths occurred from hitting pipes (instant death) rather than losing all hearts to enemies.
  - This suggests that the environment is currently a much higher threat than the combatants.
- **Mechanical Weight:**
  - Players frequently noted that the bird felt "heavy" and suggested that gravity values were too high for the current lift mechanics.
- **Feedback Deficit:**
  - There was notable confusion regarding combat outcomes; players specifically requested visual or audio feedback to confirm when a bullet successfully collided with an enemy.
- **Coordination Difficulty:**
  - Verbal feedback highlighted that the "flap and shoot" mechanism is "really difficult at first," indicating a high barrier to entry for the game's core loop.

Recurring Patterns

- **Difficulty Scaling Visibility:**
  - The implemented speed increments were described as "smooth" and "not noticeable," though this was partly because many sessions ended before the progression became significant.
- **Spatial Constraints:**
  - Players consistently suggested that pipe spacing needs to be increased to accommodate the current gravity and movement physics.
- **Combat Accuracy:**

    - Players currently have a combat efficiency of approximately 40% (4 hits out of 10 bullets fired), confirming that aiming while maintaining altitude is a significant challenge.

8 Identified Problems

1. Environmental Lethality vs. Combat Relevance

- **Problem:** There is a significant imbalance between environmental hazards and enemy interaction.
- **Evidence:** Playtest data revealed that 100% of player deaths resulted from pipe collisions (instant death) rather than heart loss from enemy encounters. This renders the health system and heart power-ups secondary to basic navigation.

2. High Barrier to Entry (Coordination Difficulty)

- **Problem:** The simultaneous requirement to manage vertical flight and horizontal combat is overly taxing for new players.
- **Evidence:** Initial survival times were as low as 5 seconds, with participants explicitly stating that the "flap and shoot" mechanism was "really difficult at first".

3. Physics and Spatial Balancing

- **Problem:** The combination of heavy gravity and tight obstacle spacing limits player maneuverability.
- **Evidence:** Multiple testers commented that the "bird felt a little heavy" and specifically recommended that "pipe spacing could be bigger" to accommodate the current gravity and mass settings.

4. Lack of Combat Sensory Feedback

- **Problem:** Players lack clear confirmation of successful actions during combat.
- **Evidence:** Testers reported confusion regarding whether projectiles hit their targets and suggested the addition of visual or audio feedback for bullet collisions and enemy destruction.

5. Difficulty Progression Visibility

- **Problem:** The speed-based difficulty scaling is too subtle to be noticed by the average player during standard play sessions.
- **Evidence:** Qualitative feedback described the speed increments as "not noticeable,"

partly because the high lethality of pipes meant sessions ended before the scaling became significant.

9 Analysis and Interpretation

Environmental Imbalance and Mechanical Weight

- **Root Cause:** The excessive difficulty of environmental navigation stems from a mismatch between the physics constants established in Sprint 2 and the procedural generation logic.
- **Analysis:** Because the “Heavy” gravity was designed before complex obstacle patterns were added, players now find it difficult to maneuver through the current pipe spacing. This creates a “frustration gap” where navigation is so demanding that players cannot utilize the health system, as any minor error results in instant environmental death rather than a recoverable heart loss.

The “Coordination Tax” on Combat

- **Root Cause:** The high barrier to entry is likely due to the implementation of the attack mechanic as a separate input (Space key) added to the primary movement input (Mouse click) in Sprint 3.
- **Analysis:** For new players, managing two distinct input types while fighting heavy physics creates a high cognitive load. The lack of sensory feedback (visual or audio) further exacerbates this; players aren't sure if their “investment” in the difficult shooting mechanic is actually paying off, leading them to revert to safe avoidance tactics.

Subtle Difficulty Scaling

- **Root Cause:** The difficulty scaling implemented in Sprint 4 was designed to be gradual to ensure a “smooth” experience, but it was tested against a baseline lethality that is currently too high.
- **Analysis:** Because players are currently dying primarily from pipes within the first 20 seconds of play, they rarely survive long enough to reach the thresholds where the speed increases become impactful. The "smoothness" of the increment essentially renders it invisible because the environment terminates the session before the challenge can meaningfully evolve.

Skill Level Skew

- **Root Cause:** The testing pool consisted entirely of software engineering students who are experienced gamers.

- **Analysis:** Even with this high-skill demographic, initial survival times were extremely low. This indicates that for your target audience (less experienced players), the current “Heavy” physics and instant-death pipes would likely be a significant deterrent to continued play.

10 Proposed Design Changes

High Priority (Immediate Action Required)

- **Physics Recalibration:** Reduce the gravity constant or increase the "flap" lift force to address player feedback that the bird feels too "heavy".
- **Combat Feedback Implementation:** Add immediate visual effects (e.g., small explosions or flashes) and audio cues when bullets hit enemies to resolve player confusion during combat.

Medium Priority (Balance and UX)

- **Adding Collectable:** Adding shield collectible to help players not die quickly and be able to be immune to enemies attacking them.

Low Priority (Future Polish)

- **Asset Finalization:** Replace current placeholder sprites with finalized art to improve visual clarity and professional feel.

11 Ethical Considerations

To ensure the playtest for Flap-and-Fight was conducted responsibly, the following ethical standards were maintained:

- **Informed Consent**: All three participants were briefed on the purpose of the test and how their gameplay data and verbal feedback would be used to improve the game.
- **Voluntary Participation**: Participants were fellow students who chose to engage in the study voluntarily and were informed they could end their play session at any time.
- **Respectful Environment**: Testers were encouraged to provide honest, unbiased feedback, whether positive or negative, without pressure from the developers.
- **Freedom to communicate**: players were able to give feedback at any moment in the gameplay test.
- **Data Privacy**: No sensitive personal information was recorded; data collection focused solely on gameplay metrics like survival time, scores, and mechanics-based observations.

Key Takeaways:

- **Mechanical Overload:** Introducing two distinct inputs (flapping and shooting) simultaneously created a high cognitive load that even experienced players struggled to master quickly.
- **Balancing Hazards:** The “instant death” mechanic of the environment made the health and heart power-up systems feel secondary, as players rarely had the chance to use them.
- **Feedback Importance:** The absence of visual and audio cues for combat results in player confusion, highlighting that mechanics are only as effective as the feedback they provide.

Future Improvements to the Testing Process:

- **Diverse Participant Pool:** Future tests should include players outside of software engineering to better represent the intended general audience and identify broader usability issues.
- **Iterative Small-Scale Testing:** To avoid the “frustration gap,” smaller, more frequent tests focusing on single mechanics (like just gravity or just combat) should occur before combining them into a full build.
- **Targeted Feedback Prompts:** While open-ended play was valuable, specifically prompting players to comment on the “weight” of the bird earlier could have pinpointed physics issues sooner in the sprint cycle.

12 Reflection

The testing process provided critical insights into the discrepancy between developer intent and actual player experience. While the team successfully implemented a complex array of features including combat and power-ups, playtesting revealed that these additions are currently overshadowed by difficult environmental physics with a small learning curve.

Additional reflections on the testing process include:

- **Cognitive Load Balance**: The “coordination tax” of managing simultaneous movement and combat was underestimated during development. Even experienced players required multiple attempts to manage the high cognitive load of flapping and shooting together.
- **System Synergy**: While individual features like hearts and shields were successfully built, they currently lack synergy with the environment due to the "instant death" pipe mechanic. This playtest highlighted that navigation difficulty must be balanced for secondary systems to remain relevant.
- **Sensory Communication**: We learned that a mechanic's technical functionality is

secondary to its visual and auditory communication. Without immediate hit feedback, players felt disconnected from the combat system, which discouraged its use.

- **Testing Demographics**: Relying on software engineering peers provided high-quality technical feedback but likely masked even greater accessibility barriers for casual players. Future iterations must account for this skill level skew to ensure the game is broadly playable.

13 Conclusion

The playtest for Flap-and-Fight (Version 2) successfully achieved its primary goals of uncovering hidden technical issues and gathering unbiased player feedback. While the core gameplay systems, including vertical movement, combat mechanics, and the scoring system, are technically functional, the test revealed a critical imbalance in game difficulty and sensory feedback.

The objective data and qualitative observations provided immense value by highlighting that the “heavy” physics and “instant death” environmental hazards currently act as a barrier to players engaging with the new combat and health features. By identifying these friction points early, the team now has a clear roadmap for prioritizing design changes, such as recalibrating physics and enhancing combat feedback, to ensure the final product is both accessible and rewarding for its target audience.